

Boundary Information Geometry for S-box

Side-Channel Leakage: Walsh Covariance, Active

Fisher Structure, and CPA Belief Dynamics

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Abstract

Side-channel attacks on lightweight block ciphers often reduce locally to distinguishing key hypotheses through leakage generated by bijective S-boxes. This paper develops an information-geometric model for these finite key-induced distributions while separating three security-relevant spaces: the singular distributions induced by exact key hypotheses, the regular full-support exponential family used for Fisher geometry, and the adversary's posterior belief simplex. For an s -bit bijective S-box S and key hypothesis k , the exact distribution $P_k(a, b) = 2^{-s} \mathbf{1}[b = S(a \oplus k)]$ is supported on one graph inside $\{0, 1\}^{2s}$ and therefore lies on the boundary of the probability simplex. Walsh characters give the key-translation rule $\hat{P}_k(\alpha, \beta) = (-1)^{\alpha \cdot k} \hat{P}_0(\alpha, \beta)$. Using this rule as a structural input, we derive closed-form boundary covariance matrices and prove that, for every bijective S-box, their active subspace has dimension $2^s - 1$ and active eigenvalues equal 2^s in the unnormalised Walsh basis. We then interpret correlation power analysis under Gaussian leakage as Bayesian dynamics on the key-belief simplex. The Hamming-weight variance $s/(4\sigma^2)$ is a bijectivity-driven baseline, whereas pairwise likelihood gaps determine key-hypothesis discrimination. Reproducible computations for PRESENT, GIFT, and SKINNY illustrate invariants, pairwise leakage gaps, and the projected active-curvature diagnostic.

Keywords: cybersecurity, side-channel analysis, correlation power analysis, S-box, information geometry, Fisher–Rao metric, Walsh–Hadamard transform, Bayesian inference

1 Introduction

1.1 Motivation

Side-channel attacks exploit implementation leakage rather than weaknesses in the abstract cipher alone. In many attacks on lightweight block ciphers, the local statistical object is a keyed S-box computation: an adversary observes or controls an input nibble or byte, measures a noisy leakage trace, and compares key hypotheses through predicted intermediate values. Correlation power analysis (CPA) is the standard example: each trace updates the adversary’s belief about the key using a leakage model such as Hamming weight.

The exact distribution induced by a keyed bijective S-box is not a smooth interior point of a probability simplex. For an s -bit bijective S-box $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$ and key hypothesis $k \in \{0, 1\}^s$, the pair (a, b) with uniform a and $b = S(a \oplus k)$ is supported on only 2^s points of the 2^{2s} -point input-output space. Hence P_k is a singular boundary distribution. This matters because ordinary Fisher–Rao tensors, geodesics, and dual-flat coordinates are defined on full-support statistical models, not directly at support-deficient endpoints.

This paper develops a boundary information-geometric framework for this side-channel setting. The central discipline is to keep separate three objects that are often conflated in informal geometric descriptions:

- (i) the regular full-support Walsh exponential family, where ordinary Fisher geometry is valid;

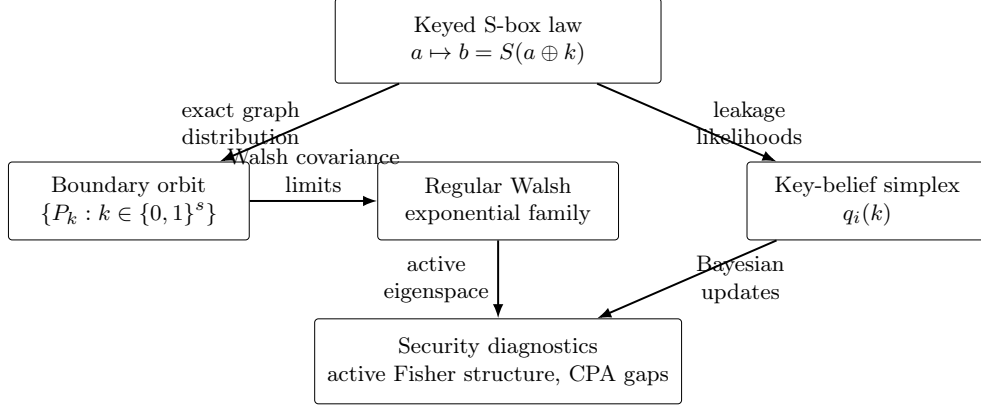


Fig. 1 Boundary-to-belief framework. The exact keyed S-box laws lie on a singular boundary orbit, the Fisher calculations are made in a regular Walsh exponential family or in boundary covariance limits, and CPA updates evolve on the separate key-belief simplex.

- (ii) the finite boundary orbit of exact key-induced S-box distributions;
- (iii) the adversary's posterior belief simplex over key hypotheses, where CPA updates take place.

This separation makes it possible to use information geometry in a controlled way: boundary distributions are represented through covariance limits, whereas Bayesian key recovery remains a regular trajectory on the belief simplex as long as the prior has full support and the leakage likelihood is positive.

1.2 Relation to prior work

The information-geometric background is the Fisher–Rao and dual-affine geometry of finite statistical models and regular exponential families [1–3]. The closest coding-theoretic precedent is the work of Ikeda, Tanaka, and Amari on LDPC and turbo codes [4]: parity-check statistics define a regular information-geometric model and decoding trajectories are interpreted through projections. Here, the sufficient statistics are Walsh–Hadamard characters on S-box input-output pairs, and the inference trajectory is the CPA posterior over key hypotheses.

On the cryptographic side, the work is related to classical Walsh/LAT and DDT analysis of vectorial Boolean functions and S-boxes, including Nyberg’s differential uniformity [5]. Four-bit S-boxes and optimal 4-bit permutations have been classified in the Boolean-function literature, notably by Leander and Poschmann [6]. The present paper does not propose a new classification of 4-bit S-boxes and does not claim that standard S-boxes with the same Walsh spectrum can be distinguished by spectral invariants. Instead, it asks a different question: what is the correct information-geometric object associated with exact keyed S-box distributions, and how should that object be connected to side-channel belief updates?

On the side-channel side, the leakage model and posterior update are consistent with differential and correlation power analysis [7, 8] and with information-theoretic analyses of key recovery attacks [9, 10]. The present framework is not a new attack and not a countermeasure. It is a formal model for separating baseline invariants forced by bijectivity from pairwise leakage gaps that control key-hypothesis discrimination.

1.3 Setting and notation

Throughout, $s \geq 1$ denotes the S-box width in bits, $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$ is a fixed bijection, $k \in \{0, 1\}^s$ is a key hypothesis, and $a \in \{0, 1\}^s$ is a uniformly distributed plaintext nibble or byte. All additions inside bit strings are over $\mathbb{F}_2$. The inner product is $\alpha \cdot a = \bigoplus_{i=1}^s \alpha_i a_i$. For $(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s$, the Walsh character is

$$\chi_{\alpha\beta}(a, b) = (-1)^{\alpha \cdot a \oplus \beta \cdot b}.$$

We write $N = 2^s$ and $\mathcal{X} = \{0, 1\}^s \times \{0, 1\}^s$.

1.4 Contributions	185
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$P_k(a, b) = 2^{-s} \mathbf{1}[b = S(a \oplus k)]$	190
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as a boundary distribution on the input-output simplex. The Walsh translation	192
rule	193
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$\hat{P}_k(\alpha, \beta) = (-1)^{\alpha \cdot k} \hat{P}_0(\alpha, \beta)$	195
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is used as a coordinate identity for the finite key orbit, not as a standalone	197
cryptanalytic novelty.	198
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in Walsh coordinates and prove that, for every bijection $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$, this	209
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forced by bijectivity and uniform inputs. The quantities that distinguish key hypotheses are the pairwise separations

$$\Delta(k, k^*) = \frac{1}{2^s} \sum_a (f(S(a \oplus k^*)) - f(S(a \oplus k)))^2,$$

which determine expected log-Bayes gains.

C5. Reproducible diagnostics on lightweight S-boxes. For PRESENT, GIFT, and SKINNY, we compute Walsh spectra, boundary covariance matrices, local spectral separation proxies, pairwise Hamming-weight leakage gaps, and the projected active-eigenspace curvature diagnostic. The examples are used to separate invariants forced by bijectivity or by the shared Walsh amplitude distribution from quantities that can vary across S-boxes.

1.5 Scope

The paper focuses on isolated bijective S-boxes under uniform inputs and on an additive Gaussian leakage model for CPA belief updates. It does not address complete multi-round cipher security, masking, higher-order leakage, non-Gaussian leakage, profiling attacks, or implementation-specific trace effects. Those topics require additional models beyond the boundary distribution and belief-update framework developed here.

2 Background

2.1 S-boxes and the Walsh–Hadamard transform

An *s-bit S-box* is a bijection $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$. The *i*-th output bit is a Boolean function $S_i : \{0, 1\}^s \rightarrow \{0, 1\}$.

Definition 2.1 (Walsh–Hadamard Transform) The Walsh–Hadamard transform (WHT) of S at $(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s$ is:

$$\widehat{S}(\alpha, \beta) = \sum_{a \in \{0, 1\}^s} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a)} \in \{-2^s, \dots, +2^s\}, \quad 2 \mid \widehat{S}(\alpha, \beta). \quad (1)$$

By convention $\widehat{S}(0, 0) = 2^s$.

The WHT encodes the two fundamental cryptographic properties of S :

- **Nonlinearity:** $\text{NL}(S) = 2^{s-1} - \frac{1}{2} \max_{(\alpha, \beta) \neq (0, 0)} |\widehat{S}(\alpha, \beta)|$.
- **Differential uniformity:** $\delta(S) = \max_{\Delta_{\text{in}} \neq 0, \Delta_{\text{out}}} |\{a : S(a) \oplus S(a \oplus \Delta_{\text{in}}) = \Delta_{\text{out}}\}|$.

The *Linear Approximation Table* (LAT) entry is $\text{LAT}_S(\alpha, \beta) = \widehat{S}(\alpha, \beta)/2$. For any S-box, Parseval’s identity gives:

$$\sum_{(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s} \widehat{S}(\alpha, \beta)^2 = 2^{2s} \cdot 2^s = 2^{3s}. \quad (2)$$

Example 2.1 (PRESENT S-box) The PRESENT-80 S-box ($s = 4$) is the hexadecimal table $S = [\text{C}, 5, 6, \text{B}, 9, 0, \text{A}, \text{D}, 3, \text{E}, \text{F}, 8, 4, 7, 1, 2]$. Its nonlinearity is $\text{NL}(S) = 4$ and differential uniformity $\delta(S) = 4$.

2.1.1 Remark on 4-bit S-box classification

The 4-bit case is small enough that strong classification results are available. In particular, optimal 4-bit S-boxes have been studied up to the standard equivalence relations used for vectorial Boolean functions [6]. Consequently, the numerical examples in Section 7 should not be interpreted as a new classification of PRESENT, GIFT, or SKINNY. Their role is narrower: they provide reproducible side-channel diagnostics for widely used lightweight S-boxes and illustrate which information-geometric quantities are forced by bijectivity or by common Walsh spectra.

323 2.2 Information geometry: essentials

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325 We collect the minimal background from Amari and Nagaoka [1] and Ay et al. [2]
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327 required in the sequel.

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330 2.2.1 Statistical manifolds and the Fisher metric

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332 Let $\mathcal{X}$ be a finite set with $|\mathcal{X}| = N$. The *probability simplex* is $\Delta_{N-1} = \{p \in \mathbb{R}^N :$
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334 $p_x \geq 0, \sum_x p_x = 1\}$. Its interior $\mathring{\Delta}_{N-1}$ is an $(N - 1)$ -dimensional manifold.

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337 **Definition 2.2** (Fisher–Rao Metric) The *Fisher–Rao metric* on $\mathring{\Delta}_{N-1}$ at p is the positive
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339 definite bilinear form

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$$341 \quad g_p(u, v) = \sum_{x \in \mathcal{X}} \frac{u_x v_x}{p_x}, \quad u, v \in T_p \mathring{\Delta}_{N-1} = \left\{ w \in \mathbb{R}^N : \sum_x w_x = 0 \right\}. \quad (3)$$

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344 By Chentsov’s theorem [3], g_F is the unique Riemannian metric on $\mathring{\Delta}_{N-1}$ (up to a
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346 positive constant) that is invariant under sufficient statistics, making it the canonical
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348 statistical distance.

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351 2.2.2 Exponential families as flat submanifolds

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353 **Definition 2.3** (Exponential Family) Given functions $T_1, \dots, T_d : \mathcal{X} \rightarrow \mathbb{R}$ (the *sufficient*
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355 *statistics*), the *exponential family* $\mathcal{E}$ is:

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$$357 \quad p(x; \theta) = \exp\left(\sum_{i=1}^d \theta_i T_i(x) - \psi(\theta)\right), \quad \theta \in \Theta \subseteq \mathbb{R}^d, \quad (4)$$

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359 where $\psi(\theta) = \log \sum_x \exp(\sum_i \theta_i T_i(x))$ is the log-partition function.

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362 The natural parameters θ_i and the expectation parameters $\eta_i = \mathbb{E}_\theta[T_i(X)]$ are
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364 related by $\eta = \nabla \psi(\theta)$ (the bijection is guaranteed when the family is minimal and
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366 regular).

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Theorem 2.1 (Dually Flat Structure, [1]) *Every regular minimal exponential family $\mathcal{E}$ is a dually flat submanifold of $\mathring{\Delta}_{N-1}$: the e -connection (exponential connection) is flat in natural parameters θ , and the m -connection (mixture connection) is flat in expectation parameters η . The generalised Pythagorean theorem holds: if $P_A, P_B, P_C \in \mathcal{E}$ and the e -geodesic from A to B is g_F -orthogonal to the m -geodesic from B to C , then*

$$D_{\text{KL}}(P_A \| P_C) = D_{\text{KL}}(P_A \| P_B) + D_{\text{KL}}(P_B \| P_C). \quad (5)$$

2.2.3 Mixture projection and Bayesian inference

Definition 2.4 (m -Projection) Given $p \in \mathring{\Delta}_{N-1}$ and a closed m -flat submanifold $\mathcal{M} \subseteq \mathring{\Delta}_{N-1}$, the m -projection of p onto $\mathcal{M}$ is:

$$\Pi_{\mathcal{M}}^{(m)}(p) = \arg \min_{q \in \mathcal{M}} D_{\text{KL}}(q \| p). \quad (6)$$

The following classical result (see [1], Chapter 3) is central to Section 5:

Proposition 2.1 (Bayesian Update as m -Projection) *Let $p(x, y) = p(x)p(y|x)$ be a joint distribution on $\mathcal{X} \times \mathcal{Y}$. After observing $y = y_0$, the posterior $p(x|y_0)$ is the m -projection of the joint p onto the submanifold $\mathcal{M}_{y_0} = \{q \in \Delta : q(y) = 0 \text{ for } y \neq y_0\}$, restricted to the x -marginal.*

2.2.4 The LDPC analogy: Ikeda-Tanaka-Amari (2004)

Ikeda, Tanaka, and Amari [4] develop a complete information-geometric framework for LDPC and turbo codes. They model the joint distribution over a binary codeword $x \in \{-1, +1\}^N$ as:

$$p(x; \theta, v) = \exp\left(\theta \cdot x + \sum_r v_r c_r(x) - \psi\right), \quad (7)$$

where $c_r(x) = \prod_{i \in r} x_i$ are parity check functions and v_r the constraint weights. They show: (i) the e/m dual structure is exact; (ii) Belief Propagation decoding corresponds to iterated e/m -projections; (iii) curvature of the statistical manifold controls decoding convergence.

In the present paper, the parity functions $c_r(x)$ are replaced by the Walsh-Hadamard characters $\chi_{\alpha\beta}(a, b)$ of the S-box, and LDPC decoding is replaced by CPA key recovery. Table 1 summarises the correspondence.

Table 1 Structural analogy between the LDPC framework of Ikeda-Tanaka-Amari (2004) and the S-box framework of this paper.

Ikeda-Tanaka-Amari (LDPC)	This paper (S-box)
Codeword $x \in \{-1, +1\}^N$	Input-output pair $(a, b) \in \{0, 1\}^{2s}$
Parity check $c_r(x) = \prod_{i \in r} x_i$	Walsh character $\chi_{\alpha\beta}(a, b) = (-1)^{\alpha \cdot a \oplus \beta \cdot b}$
Channel log-likelihood θ_i	Key log-likelihood ratio $\log(q_k/q_{k'})$
Parity constraint weight v_r	Walsh spectrum value $\hat{S}(\alpha, \beta)$
Belief Propagation decoding	CPA key recovery
Decoding convergence rate	Per-trace Fisher information $I_S(\sigma)$

3 The S-box Statistical Model

3.1 Key-induced boundary distributions

Fix a bijective S-box $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$ and let A be uniformly distributed over $\{0, 1\}^s$. Under key hypothesis $k \in \{0, 1\}^s$, the observed input-output pair is $(A, S(A \oplus k))$. This induces the following distribution on $\mathcal{X} = \{0, 1\}^s \times \{0, 1\}^s$.

Definition 3.1 (Key-induced distribution) For $k \in \{0, 1\}^s$, define

$$P_k(a, b) = 2^{-s} \mathbf{1}[b = S(a \oplus k)]. \quad (8)$$

The support of P_k is the graph $G_k = \{(a, S(a \oplus k)) : a \in \{0, 1\}^s\}$.

Each P_k is a boundary point of the simplex $\Delta_{2^{2s}-1}$, because it assigns zero mass to $2^{2s} - 2^s$ outcomes. This support deficiency is central: ordinary Fisher–Rao tensors of the full simplex are defined on the interior, not directly at P_k . We therefore treat P_k as a singular endpoint of an interior regularization when differential-geometric operations are needed.

3.2 Walsh representation and key translation

Definition 3.2 (Walsh coefficient of P_k) For $(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s$, set

$$\hat{P}_k(\alpha, \beta) = \sum_{(a,b) \in \mathcal{X}} \chi_{\alpha\beta}(a,b) P_k(a,b) = 2^{-s} \sum_{a \in \{0,1\}^s} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a \oplus k)}. \quad (9)$$

Lemma 3.1 (Walsh key-translation rule) For every $k \in \{0, 1\}^s$ and every $(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s$,

$$\hat{P}_k(\alpha, \beta) = (-1)^{\alpha \cdot k} \hat{P}_0(\alpha, \beta) = 2^{-s} (-1)^{\alpha \cdot k} \hat{S}(\alpha, \beta). \quad (10)$$

Proof Using $a' = a \oplus k$,

$$\hat{P}_k(\alpha, \beta) = 2^{-s} \sum_{a'} (-1)^{\alpha \cdot (a' \oplus k) \oplus \beta \cdot S(a')} = 2^{-s} (-1)^{\alpha \cdot k} \sum_{a'} (-1)^{\alpha \cdot a' \oplus \beta \cdot S(a')},$$

which is exactly the claimed identity. $\square$

Remark 3.1 (Walsh orbit) The key does not change the amplitude $|\hat{S}(\alpha, \beta)|$; it only changes the sign according to the character $\alpha \mapsto (-1)^{\alpha \cdot k}$. Thus $\{P_k : k \in \{0, 1\}^s\}$ is best described as a finite Walsh-phase orbit, not as a smooth manifold. The rule is a structural translation identity; the information-geometric content in the sequel comes from the covariance limit, active subspace, and belief-update geometry built on top of this identity.

507 3.3 The regular Walsh exponential family

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509 **Definition 3.3** (Walsh exponential family) Let $T_{\alpha\beta} = \chi_{\alpha\beta}$ for $(\alpha, \beta) \neq (0, 0)$. The Walsh
 510 exponential family on $\mathcal{X}$ is

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$$512 \quad p_{\theta}(a, b) = \exp \left(\sum_{(\alpha, \beta) \neq (0, 0)} \theta_{\alpha\beta} T_{\alpha\beta}(a, b) - \psi(\theta) \right), \quad \theta \in \mathbb{R}^{2^{2s}-1}. \quad (11)$$

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 516 Because the nonconstant Walsh characters together with the constant character
 517 form an orthogonal basis of all real functions on $\mathcal{X}$, this is simply the full regular
 518 exponential family over $\mathcal{X}$ written in Walsh coordinates. Its image is the interior of
 519 the probability simplex. The singular distributions P_k belong to the closure, not to
 520 the regular parameter space.
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 526 **Proposition 3.1** (Boundary expectation coordinates) *For each k and $(\alpha, \beta) \neq (0, 0)$,*

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$$528 \quad \eta_{\alpha\beta}^{(k)} := \mathbb{E}_{P_k}[T_{\alpha\beta}] = (-1)^{\alpha \cdot k} 2^{-s} \widehat{S}(\alpha, \beta). \quad (12)$$

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530 *The corresponding natural parameters are not finite; they are obtained only as limits of*
 531 *full-support approximations approaching P_k .*

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 535 *Proof* The first equality is the definition of expectation coordinates and the second is
 536 Lemma 3.1. Since P_k has zeros on $\mathcal{X}$, no finite vector θ in (11) can represent it. $\square$

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540 3.4 Interior regularization and boundary covariance form

Let U denote the uniform distribution on $\mathcal{X}$. For $\varepsilon \in (0, 1)$ define

$$P_k^{\varepsilon} = (1 - \varepsilon)P_k + \varepsilon U. \quad (13)$$

Then P_k^{ε} has full support and belongs to the regular simplex interior. Fisher–Rao
 calculations are unambiguous at P_k^{ε} . The finite covariance limit as $\varepsilon \downarrow 0$ is the following.

Definition 3.4 (Boundary Fisher covariance form) The boundary covariance form associated with key k is

$$\mathcal{I}_{ij}^\partial(k) = \text{Cov}_{P_k}(T_i, T_j), \quad i, j \in \{0, 1\}^s \times \{0, 1\}^s \setminus \{(0, 0)\}. \quad (14)$$

It is positive semidefinite and generally singular. We do not call it a non-degenerate Fisher metric at P_k .

Proposition 3.2 (Closed form of the boundary covariance) For $i = (\alpha, \beta)$ and $j = (\alpha', \beta')$,

$$\begin{aligned} \mathcal{I}_{ij}^\partial(k) &= \text{Cov}_{P_k}(T_{\alpha\beta}, T_{\alpha'\beta'}) \\ &= (-1)^{(\alpha \oplus \alpha') \cdot k} \left(2^{-s} \widehat{S}(\alpha \oplus \alpha', \beta \oplus \beta') - 2^{-2s} \widehat{S}(\alpha, \beta) \widehat{S}(\alpha', \beta') \right). \end{aligned} \quad (15)$$

In particular,

$$\mathcal{I}_{(\alpha\beta), (\alpha\beta)}^\partial(k) = 1 - \left(2^{-s} \widehat{S}(\alpha, \beta) \right)^2. \quad (16)$$

Proof Since $T_i T_j = T_{i \oplus j}$,

$$\mathbb{E}_{P_k}[T_i T_j] = \widehat{P}_k(\alpha \oplus \alpha', \beta \oplus \beta') = (-1)^{(\alpha \oplus \alpha') \cdot k} 2^{-s} \widehat{S}(\alpha \oplus \alpha', \beta \oplus \beta').$$

Subtracting $\mathbb{E}_{P_k}[T_i] \mathbb{E}_{P_k}[T_j]$ and applying Lemma 3.1 gives (15). The diagonal follows from $T_i^2 = 1$. $\square$

Lemma 3.2 (Key covariance isometry) Let D_k be the diagonal sign matrix $(D_k)_{(\alpha\beta), (\alpha\beta)} = (-1)^{\alpha \cdot k}$. Then

$$\mathcal{I}^\partial(k) = D_k \mathcal{I}^\partial(0) D_k. \quad (17)$$

Thus every spectral invariant of the boundary covariance form is independent of the true key.

Proof Equation (15) factors exactly into the sign contribution $(-1)^{\alpha \cdot k} (-1)^{\alpha' \cdot k}$ times the $k = 0$ covariance entry. $\square$

Remark 3.2 (What is metric and what is not) For $\varepsilon > 0$, the Fisher–Rao metric at P_k^ε is an ordinary Riemannian metric. At $\varepsilon = 0$, only the finite covariance form (14) is used. Distances or curvatures involving the exact P_k are therefore either (i) computed on the belief simplex, (ii) computed for P_k^ε and then studied as limits, or (iii) explicitly described as first-order/local covariance approximations.

3.5 First-order spectral separation of key hypotheses

The exact distributions P_k and $P_{k'}$ have disjoint supports when $k \neq k'$, so their KL divergence is infinite. Nevertheless, the Walsh coordinate displacement has a simple closed form:

$$\eta_{\alpha\beta}^{(k')} - \eta_{\alpha\beta}^{(k)} = \left((-1)^{\alpha \cdot k'} - (-1)^{\alpha \cdot k} \right) 2^{-s} \widehat{S}(\alpha, \beta). \quad (18)$$

Consequently, only Walsh modes satisfying $\alpha \cdot (k \oplus k') = 1$ contribute to the spectral separation. A diagonal local approximation based on the boundary covariance form is

$$d_{\text{loc}}^2(k, k') = \sum_{\substack{(\alpha, \beta) \neq (0, 0): \\ \alpha \cdot (k \oplus k') = 1}} \frac{4 \cdot 2^{-2s} \widehat{S}(\alpha, \beta)^2}{1 - 2^{-2s} \widehat{S}(\alpha, \beta)^2}, \quad (19)$$

provided the denominator is nonzero. This quantity is a local spectral proxy, not the exact Fisher–Rao geodesic distance between the singular distributions.

4 Dual Structure, Boundary Limits, and Belief

Geometry

4.1 Regular dual flatness of the interior family

Theorem 4.1 (Regular dual flatness) *The Walsh exponential family (11) is a regular minimal exponential family on $\mathcal{X}$. Its interior carries the standard dually flat information-geometric structure: natural coordinates θ are e -affine, expectation coordinates η are m -affine, the log-partition function ψ is the e -potential, and its Legendre dual is $\varphi(\eta) = -H(p_\eta)$.*

Proof The natural parameter space is the open set $\mathbb{R}^{2^s-1}$. The sufficient statistics are bounded, so the partition function is finite and smooth for all θ . Minimality follows from the orthogonality and linear independence of the nonconstant Walsh characters together with the constant character. The standard theorem for regular minimal exponential families then gives dual flatness and the Legendre correspondence [1, 2]. $\square$

Remark 4.1 (Boundary discipline) Theorem 4.1 applies to full-support distributions only. The exact key distributions P_k lie in the closure. Statements about geodesics, Christoffel symbols, Bregman identities, or curvature at P_k are therefore meaningful only after regularization or after being rephrased as finite boundary covariance statements.

4.2 KL divergence between exact key distributions

Proposition 4.1 (Support separation) *If S is bijective and $k \neq k'$, then*

$$\text{supp}(P_k) \cap \text{supp}(P_{k'}) = \emptyset, \quad D_{\text{KL}}(P_k \| P_{k'}) = +\infty. \quad (20)$$

Proof If (a, b) belongs to both supports, then $S(a \oplus k) = b = S(a \oplus k')$. Since S is injective, $a \oplus k = a \oplus k'$, hence $k = k'$, a contradiction. Thus $P_{k'}$ assigns zero probability to every point in the support of P_k when $k \neq k'$, which implies infinite KL divergence. $\square$

Remark 4.2 (No finite Bregman identity at singular endpoints) The usual equality between KL divergence and the Bregman divergence of the dual potential is an interior statement. For P_k and $P_{k'}$ with disjoint supports, one may study $D_{\text{KL}}(P_k^\varepsilon \| P_{k'}^\varepsilon)$ and then let $\varepsilon \downarrow 0$, but the limiting divergence is infinite. We therefore do not use finite Pythagorean decompositions directly between distinct exact key distributions.

691 4.3 Mixtures induced by key beliefs

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693 Let $Q = \Delta_{2^s-1}^\circ$ be the interior of the simplex over keys. A belief $q \in Q$ induces a
694 distribution on input-output pairs

$$695 \quad \bar{P}_q(a, b) = \sum_{k \in \{0,1\}^s} q(k) P_k(a, b) = 2^{-s} \sum_{k \in \{0,1\}^s} q(k) \mathbf{1}[b = S(a \oplus k)]. \quad (21)$$

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702 If q has full support, then $\bar{P}_q$ has full support on $\mathcal{X}$.

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704
705 **Proposition 4.2** (Walsh representation of belief mixtures) *Let*

$$706 \quad \hat{q}(\alpha) = \sum_{k \in \{0,1\}^s} q(k) (-1)^{\alpha \cdot k}.$$

707
708
709 *Then*

$$710 \quad \bar{P}_q(a, b) = 2^{-2s} \sum_{\alpha, \beta \in \{0,1\}^s} \hat{q}(\alpha) \hat{S}(\alpha, \beta) (-1)^{\alpha \cdot a \oplus \beta \cdot b}. \quad (22)$$

711
712
713 *In particular, the uniform belief induces the uniform distribution on $\mathcal{X}$.*

714
715
716
717
718
719
720
721 *Proof* Apply Walsh inversion to the indicator $\mathbf{1}[b = S(a \oplus k)]$ and sum over k with weights
722 $q(k)$. For uniform q , $\hat{q}(\alpha) = \mathbf{1}[\alpha = 0]$, and $\hat{S}(0, 0) = 2^s$ while $\hat{S}(0, \beta) = 0$ for $\beta \neq 0$ because S
723 is bijective. □

724 4.4 Fisher–Rao geometry of the belief simplex

725
726
727
728
729 **Definition 4.1** (Belief simplex metric) For $q \in Q$ and tangent vectors u, v with $\sum_k u_k =$
730 $\sum_k v_k = 0$, the Fisher–Rao metric is

$$731 \quad g_q^{(Q)}(u, v) = \sum_{k \in \{0,1\}^s} \frac{u_k v_k}{q(k)}. \quad (23)$$

Under the square-root map $q \mapsto (\sqrt{q(k)})_k$, this metric is the round metric on a sphere of radius 2. Therefore

$$d_F(q, q') = 2 \arccos \left(\sum_{k \in \{0,1\}^s} \sqrt{q(k)q'(k)} \right). \quad (24)$$

For distinct point-mass limits, $d_F(\delta_k, \delta_{k'}) = \pi$. This is a statement about the belief simplex, not about geodesics between the singular input-output distributions P_k and $P_{k'}$.

4.5 Pythagorean identities

The generalized Pythagorean theorem is used in the following restricted sense. For full-support distributions in the regular Walsh exponential family, the standard identity

$$D_{\text{KL}}(P_A \| P_C) = D_{\text{KL}}(P_A \| P_B) + D_{\text{KL}}(P_B \| P_C) \quad (25)$$

holds whenever the e-geodesic from A to B is orthogonal to the m-geodesic from B to C . In the CPA setting, the relevant full-support object is the posterior belief $q_n \in Q$ as long as the prior has full support and the Gaussian likelihood is strictly positive. This is why projection statements in Section 5 are made on Q or on regularized joint distributions, not directly on disjoint-support P_k endpoints.

783 5 Correlation Power Analysis as Bayesian Projection 784 785 Dynamics 786

787 5.1 Leakage model 788 789

790 We consider the standard non-profiled CPA setting. The adversary knows S , observes
791 plaintexts $a_1, \dots, a_n$ uniformly drawn from $\{0, 1\}^s$, and obtains leakage measurements
792

$$793 \quad \quad \quad 794 \quad \quad \quad 795 \quad \quad \quad W_i = f(S(a_i \oplus k^*)) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad (26)$$

796
797
798 where k^* is the true key and $f : \{0, 1\}^s \rightarrow \mathbb{R}$ is the leakage model. In the numerical
799 experiments we take $f = \text{HW}$.
800
801

802 5.2 Bayesian update as a mixture projection 803 804

805 **Theorem 5.1** (CPA posterior update as m -projection) *Let $q_{i-1} \in \Delta_{2^s-1}^\circ$ be the prior belief*
806 *over keys before observing (a_i, W_i) . The posterior*
807

$$808 \quad \quad \quad 809 \quad \quad \quad q_i(k) = \frac{q_{i-1}(k) p(W_i | a_i, k)}{\sum_{h \in \{0, 1\}^s} q_{i-1}(h) p(W_i | a_i, h)} \quad (27)$$

810
811 *is the m -projection of the joint prior-likelihood distribution onto the affine constraint defined*
812 *by the observation, marginalised to the key coordinate.*
813
814

815
816 *Proof* The joint distribution before conditioning is $q_{i-1}(k)p(a_i)p(W_i | a_i, k)$. Conditioning on
817 the observed value imposes an affine constraint. Minimising the Kullback-Leibler divergence
818 under that constraint yields the conditional distribution, whose key marginal is exactly (27).
819
820 □

821
822 The exact S-box distributions P_k lie on the boundary of the simplex, but the belief
823 trajectory $q_0, q_1, \dots, q_n$ stays in the interior of the belief simplex as long as the prior
824 has full support and the Gaussian likelihood is positive. Hence CPA can be studied as
825 a regular trajectory on $(Q, g_F^{(Q)})$.
826
827
828

5.3 Per-trace signal information

Definition 5.1 (Per-trace signal information) For a leakage model f and noise variance σ^2 , define

$$I_S(f, \sigma) = \sigma^{-2} \text{Var}_{a \sim \text{Unif}(\{0,1\}^s)} [f(S(a \oplus k^*))]. \quad (28)$$

When S is bijective, this quantity does not depend on k^* .

Proposition 5.1 (Hamming-weight specialization) For $f = \text{HW}$ and a bijective S ,

$$I_S(\text{HW}, \sigma) = \frac{s}{4\sigma^2}. \quad (29)$$

In particular, for 4-bit bijective S -boxes, $I_S(\text{HW}, \sigma) = 1/\sigma^2$.

Proof Bijectivity of S together with the uniformity of $a \oplus k^*$ implies that $S(a \oplus k^*)$ is uniform over $\{0,1\}^s$. The Hamming weight of a uniform s -bit word follows a binomial distribution $\text{Binomial}(s, 1/2)$, whose variance is $s/4$. $\square$

5.4 Pairwise log-likelihood increments

For a competing key $k \neq k^*$, the per-trace squared separation is

$$\Delta(k, k^*) = \frac{1}{2^s} \sum_{a \in \{0,1\}^s} (f(S(a \oplus k^*)) - f(S(a \oplus k)))^2. \quad (30)$$

Proposition 5.2 (Expected pairwise log-Bayes gain) Under the Gaussian leakage model,

$$\mathbb{E} \left[\log \frac{p(W | A, k^*)}{p(W | A, k)} \right] = \frac{\Delta(k, k^*)}{2\sigma^2}, \quad (31)$$

where the expectation is over A (uniform) and W conditioned on A and k^* .

Proof For a fixed plaintext a , the two likelihoods are Gaussian with the same variance σ^2 and means $\mu_{k^*}(a) = f(S(a \oplus k^*))$, $\mu_k(a) = f(S(a \oplus k))$. The Kullback-Leibler divergence between

875 two univariate Gaussians with equal variance is $(\mu_*(a) - \mu_k(a))^2/(2\sigma^2)$. Averaging over a
876 gives the result. □

878

879 The worst-case pairwise gap

880

881

$$882 \quad \Gamma_S(f, \sigma) = \min_{k \neq k^*} \frac{\Delta(k, k^*)}{2\sigma^2} \quad (32)$$

883

884

885 provides a rigorous likelihood-separation measure, distinct from $I_S(f, \sigma)$ in general.

886

887

888 5.5 Model-based sample-complexity estimate

889

890 Starting from a uniform prior, the initial key uncertainty equals $s \log 2$ nats. A first-order
891 information estimate then yields

892

893

$$894 \quad n_{\text{est}} \approx \frac{s \log 2}{I_S(f, \sigma)}. \quad (33)$$

895

896

897 For 4-bit Hamming-weight leakage, $n_{\text{est}} \approx 4 \log 2 \sigma^2$. This is a heuristic leakage-rate
898 estimate. A more rigorous bound follows from the pairwise gap $\Gamma_S(f, \sigma)$: the expected
899 log-posterior odds against any fixed wrong key grow at rate at least $\Gamma_S(f, \sigma)$.

900

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902 6 Active-Eigenspace Curvature: Computation and

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904 Status

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907 6.1 What curvature is being computed

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909 The regular Walsh exponential family is dually flat with respect to the e- and m-
910 connections, but its Levi-Civita curvature is not automatically zero. At exact key
911 distributions P_k , however, the ordinary Riemannian tensor of the full-support family
912 is not directly defined because P_k lies on the boundary. This section therefore uses the
913 following restricted object.

914

Definition 6.1 (Active covariance eigenspace) Let $\mathcal{I}^\partial(k)$ be the boundary covariance matrix in Definition 3.4. Its active eigenspace is the span of the eigenvectors with strictly positive eigenvalues. Curvature computations below are performed after projecting the covariance and third cumulant tensors to this active eigenspace.

The resulting values are curvature diagnostics of the limiting regularized geometry obtained from the active covariance eigenspace. The finite orbit $\{P_k\}$ itself is discrete and is used here through its induced boundary covariance and cumulant tensors.

6.2 Universal rank and eigenvalues of the boundary covariance

Theorem 6.1 (Rank and active eigenvalues) *For any bijection $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$, the boundary covariance matrix $\mathcal{I}^\partial(k)$ has rank $2^s - 1$. Its nonzero eigenvalues are all 2^s when the feature coordinates are the unnormalised nonconstant Walsh characters used in Definition 3.3.*

Proof Let $N = 2^s$ and index rows by $a \in \{0, 1\}^s$. Let X be the $N \times (2^{2s} - 1)$ feature matrix with entries $X_{a,(\alpha,\beta)} = (-1)^{\alpha \cdot a \oplus \beta \cdot S(a \oplus k)}$ for $(\alpha, \beta) \neq (0, 0)$. The boundary covariance is $N^{-1}X^\top HX$, where $H = I_N - N^{-1}\mathbf{1}\mathbf{1}^\top$ centers over the support graph. The nonzero eigenvalues of $N^{-1}X^\top HX$ equal those of $N^{-1}HXX^\top H$. For $a, a' \in \{0, 1\}^s$,

$$(XX^\top)_{a,a'} = \sum_{(\alpha,\beta) \neq (0,0)} (-1)^{\alpha \cdot (a \oplus a') \oplus \beta \cdot (S(a \oplus k) \oplus S(a' \oplus k))} = N^2 \mathbf{1}[a = a'] - 1,$$

because S is injective. Hence $N^{-1}HXX^\top H = N^{-1}H(N^2I_N - \mathbf{1}\mathbf{1}^\top)H = NH$. The matrix H has rank $N - 1$ and eigenvalue 1 on the centered subspace. Therefore the active eigenvalues are all $N = 2^s$. $\square$

6.3 Third cumulants and curvature protocol

Let

$$C_{ijk}^\partial(k) = \mathbb{E}_{P_k} [(T_i - \eta_i)(T_j - \eta_j)(T_k - \eta_k)] \quad (34)$$

967 be the boundary third cumulant tensor. Since products of Walsh characters are again
 968 Walsh characters, every entry can be computed exactly from the WHT of S and the
 969 key-translation rule. If U is an orthonormal basis of the active eigenspace and $\lambda_a = 2^s$
 970 are the active covariance eigenvalues, the projected tensor is
 971

$$\tilde{C}_{abc} = \sum_{i,j,\ell} U_{ia} U_{jb} U_{\ell c} C_{ij\ell}^\partial. \quad (35)$$

972 The Levi-Civita curvature diagnostic used in the experiments is then computed with
 973 the standard dually-flat formula

$$R_{abab} = \frac{1}{4} \sum_c \frac{\tilde{C}_{aac} \tilde{C}_{bbc} - \tilde{C}_{abc}^2}{\lambda_c}, \quad K_{ab} = \frac{R_{abab}}{\lambda_a \lambda_b}, \quad a < b. \quad (36)$$

974 Both verification scripts implement Equation (36) directly.

975 6.4 Empirical constant-curvature finding

976 **Proposition 6.1** (Reproducible 4-bit curvature observation) *For the PRESENT, GIFT, and*
 977 *SKINNY 4-bit S-boxes, the projected active-eigenspace curvature diagnostic (36) gives*

$$K_{ab} = -\frac{1}{4} \quad (37)$$

978 for all $\binom{15}{2} = 105$ active coordinate planes, up to numerical precision in the provided scripts.

979 *Verification protocol* The computation is reproducible from the repository scripts. The scripts
 980 build the WHT table, construct $\mathcal{I}^\partial$, diagonalize its active 15-dimensional eigenspace, project
 981 the third cumulant tensor, and evaluate (36). The reported value is stable across the three
 982 tested S-boxes and an additional random 4-bit bijection in `verify_curvature.py`.

983 These reproducible computations suggest the following conjecture: for bijective S-boxes,
 984 the active Fisher subspace associated with the Walsh boundary covariance form has constant
 985 sectional curvature $K = -1/4$, under the curvature convention and projection procedure used

in this work. A symbolic proof, or a characterization of the precise hypotheses under which the conjecture holds, is left as an open problem. $\square$

6.5 Relation to the belief simplex

The belief simplex $Q = \Delta_{2^s-1}^\circ$ with Fisher–Rao metric is isometric, via the square-root embedding, to a sphere of radius 2 and hence has constant sectional curvature $+1/4$. The observed value $-1/4$ in the active boundary covariance diagnostic suggests a sign-dual phenomenon between the posterior belief geometry and the limiting keyed S-box covariance geometry, which is recorded here as part of the open geometric structure of the model.

7 Reproducible Security Diagnostics

The computations in this section are not intended to classify 4-bit S-boxes or to benchmark full cipher attacks. They serve two narrower purposes. First, they verify the exact boundary covariance and active-eigenspace formulas on standard lightweight S-boxes. Second, they separate quantities that are structural invariants from quantities that can influence CPA discrimination between key hypotheses.

7.1 Target S-boxes

We evaluate the 4-bit bijective S-boxes used in PRESENT, GIFT, and SKINNY:

$$S_{\text{PRESENT}} = [C, 5, 6, B, 9, 0, A, D, 3, E, F, 8, 4, 7, 1, 2],$$

$$S_{\text{GIFT}} = [1, A, 4, C, 6, F, 3, 9, 2, D, B, 7, 5, 0, 8, E],$$

$$S_{\text{SKINNY}} = [C, 6, 9, 0, 1, A, 2, B, 3, 8, 5, D, 4, E, 7, F].$$

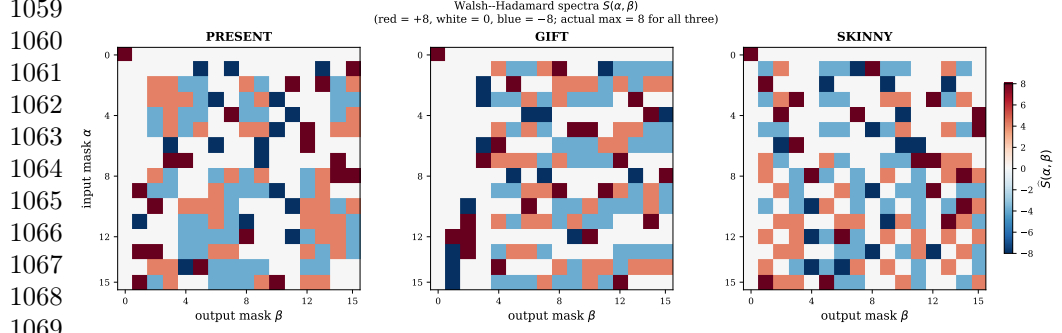


Fig. 2 Walsh spectra. The plotted Walsh tables show the same amplitude profile for PRESENT, GIFT, and SKINNY, which explains why purely amplitude-based diagnostics coincide for these examples.

These examples are useful because they are widely used in lightweight cryptography and because exhaustive enumeration over all inputs, masks, and key hypotheses is possible.

7.2 Walsh spectra and boundary covariance

For each S-box we compute the full WHT table

$$\hat{S}(\alpha, \beta) = \sum_{a \in \{0,1\}^4} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a)}.$$

The three tested S-boxes have the same absolute Walsh-coefficient distribution over nonzero masks:

$$\#\|\hat{S}\| = 0 : 123, \quad \#\|\hat{S}\| = 4 : 96, \quad \#\|\hat{S}\| = 8 : 36.$$

Therefore any diagnostic depending only on the Walsh amplitudes must coincide for these three examples. This is an expected consequence of the spectral data, not a new distinguishing property.

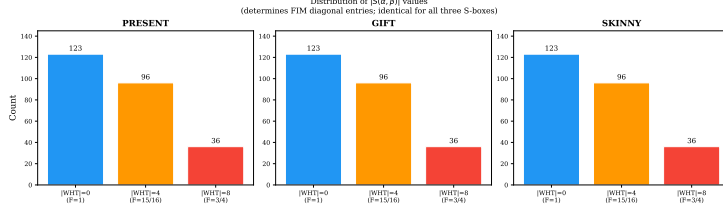


Fig. 3 Walsh amplitude distribution. The histogram records the shared counts of absolute Walsh coefficients over nonzero masks for the three tested S-boxes.

Table 2 Boundary covariance structure for the tested 4-bit bijections.

S-box	Matrix size	Rank	Active eigenvalues
PRESENT	255×255	15	16 repeated 15 times
GIFT	255×255	15	16 repeated 15 times
SKINNY	255×255	15	16 repeated 15 times

Table 3 Interpretation of the key-separation quantities.

Quantity	Status
$D_{\text{KL}}(P_k \ P_{k'})$ for $k \neq k'$	Infinite, because supports are disjoint
$d_F(\delta_k, \delta_{k'})$ on the belief simplex	Exactly π
$d_{\text{loc}}^2(k, k')$ in (19)	Local Walsh/Fisher covariance proxy

The boundary covariance matrix is built from Proposition 3.2. For $s = 4$ it is a 255×255 positive semidefinite matrix. Consistent with Theorem 6.1, all three matrices have rank 15 and active eigenvalues equal to 16 up to numerical precision.

7.3 Local spectral separation

Using (19), we compute local diagonal Walsh/Fisher separation proxies between key hypotheses. This value is not the exact Fisher–Rao distance between singular distributions. It is a local covariance-weighted coordinate displacement. The key-translation rule implies that the value depends only on the key difference $k \oplus k'$.

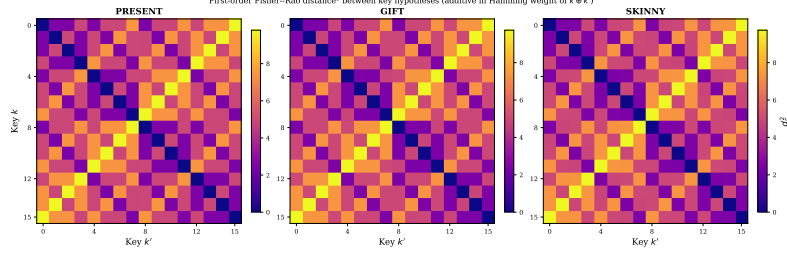


Fig. 4 Local key-separation proxy. The matrix visualises the covariance-weighted Walsh displacement between key hypotheses and should be read as a local diagnostic, not as an exact distance between singular laws.

Table 4 Pairwise Hamming-weight leakage gaps over nonzero key differences.

S-box	$\min_{\delta \neq 0} \Delta(\delta)$	$\max_{\delta \neq 0} \Delta(\delta)$	$\frac{1}{15} \sum_{\delta \neq 0} \Delta(\delta)$
PRESENT	1	7/2	32/15
GIFT	5/4	7/2	32/15
SKINNY	1	3	32/15

7.4 CPA leakage gaps

For bijective 4-bit S-boxes and uniform inputs, the scalar Hamming-weight variance in Proposition 5.1 is always

$$I_S(\text{HW}, \sigma) = \frac{1}{\sigma^2}.$$

This scalar is a baseline leakage descriptor and cannot distinguish PRESENT, GIFT, and SKINNY. Pairwise gaps

$$\Delta(\delta) = \frac{1}{16} \sum_{a \in \{0,1\}^4} (\text{HW}(S(a)) - \text{HW}(S(a \oplus \delta)))^2, \quad \delta \neq 0,$$

are more informative for CPA because the expected log-Bayes gain against a wrong key with difference δ is $\Delta(\delta)/(2\sigma^2)$.

Table 4 shows why the scalar variance descriptor should not be overinterpreted: all three S-boxes have the same average pairwise gap, but the worst and best key-difference

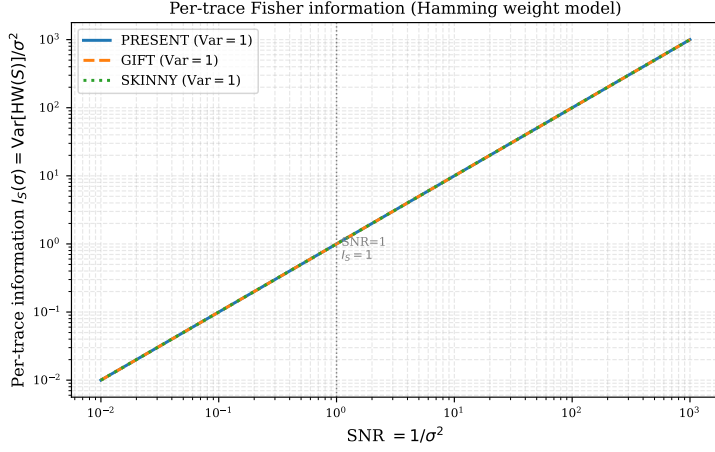


Fig. 5 Per-trace leakage baseline. The scalar Hamming-weight variance is the same for all bijective 4-bit S-boxes under uniform inputs, so CPA discrimination must be interpreted through pairwise leakage gaps rather than this baseline alone.

Table 5 Projected active-eigenspace curvature diagnostic for the tested 4-bit S-boxes.

S-box	Active dimension	Planes	Min K	Max K
PRESENT	15	105	−0.25	−0.25
GIFT	15	105	−0.25	−0.25
SKINNY	15	105	−0.25	−0.25

cases vary. These pairwise gaps are the quantities that enter expected posterior-odds growth in the Gaussian leakage model.

7.5 Active-eigenspace curvature

The curvature diagnostic of Section 6 is evaluated in the 15-dimensional active eigenspace of the boundary covariance matrix. The result is constant for all coordinate planes in the tested cases.

7.6 Reproducibility checklist

The computational claims in this section can be checked as follows:

- 1243 (i) `gen_figures.py` regenerates the Walsh-spectrum and key-separation figures in
1244 the project `figures/` directory.
1245
1246 (ii) `compute_curvatures.py` computes the active covariance eigenspace and all 105
1247 curvature values.
1248
1249 (iii) `verify_curvature.py` checks the value $K = -1/4$ for PRESENT and for an
1250 additional fixed random 4-bit bijection.
1251
1252
1253 (iv) The rank/eigenvalue claim follows independently from Theorem 6.1.
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1255
1256

1257 8 Discussion

1259 8.1 What the framework establishes

1261 The framework establishes a controlled information-geometric model for the local S-box
1262 distributions that appear in side-channel key recovery. Its first role is structural: exact
1263 key-induced distributions are singular boundary points, while ordinary Fisher–Rao
1264 geometry belongs to full-support interiors or to the adversary’s belief simplex. This
1265 distinction prevents assigning non-degenerate Fisher metrics, finite KL divergences, or
1266 ordinary geodesics to disjoint-support key endpoints.

1267 The second role is algebraic. Walsh characters give explicit coordinates for the
1268 boundary orbit, and the key-translation rule determines how expectation coordinates
1269 move when the key hypothesis changes. This rule is elementary, but it is useful because
1270 it gives closed-form covariance and displacement expressions.

1271 The third role is security-oriented. CPA is a Bayesian trajectory on the belief
1272 simplex. Under the Gaussian leakage model, pairwise Hamming-weight gaps determine
1273 expected log-Bayes gains against wrong keys. The scalar leakage variance is therefore
1274 a baseline descriptor, whereas pairwise gaps describe the discrimination structure
1275 relevant to key recovery.

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8.2 What is not claimed	1289
	1290
The paper does not provide a new classification of 4-bit S-boxes. The classification	1291
and equivalence theory of small vectorial Boolean functions is an established subject.	1292
PRESENT, GIFT, and SKINNY are used as reproducible lightweight examples, not as	1293
evidence that the proposed spectral quantities separate S-boxes with the same Walsh	1294
amplitude distribution.	1295
	1296
	1297
	1298
The paper also does not propose a new side-channel attack or countermeasure. The	1299
CPA model is the standard Gaussian Hamming-weight model. The contribution is the	1300
geometric separation of the boundary S-box law from the posterior belief law and the	1301
identification of which quantities are forced by bijectivity.	1302
	1303
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	1307
8.3 Cryptographic interpretation	1308
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	1310
For isolated bijective 4-bit S-boxes under uniform input and Hamming-weight leak-	1311
age, $I_S(\text{HW}, \sigma) = 1/\sigma^2$ for every bijection. This equality is not a ranking criterion	1312
for S-boxes. Practical side-channel behavior depends on pairwise leakage gaps, imple-	1313
mentation leakage, noise, masking, alignment, higher-order effects, and the full cipher	1314
context.	1315
	1316
	1317
	1318
The boundary covariance and local Walsh/Fisher separation proxy complement	1319
LAT, DDT, and empirical leakage evaluation. They are best understood as coordinate	1320
diagnostics for the singular key orbit and as inputs to posterior belief dynamics, not	1321
as replacements for standard cryptanalytic criteria.	1322
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	1327
8.4 Status of the curvature finding	1328
	1329
	1330
The active-eigenspace value $K = -1/4$ is reported as a reproducible diagnostic under	1331
the projection, normalization, and sign conventions specified in Section 6. The rank and	1332
active eigenvalue statement is proved for every bijective S-box. A complete symbolic	1333
	1334

1335 characterization of the curvature diagnostic, including the exact hypotheses under
1336 which it is constant, remains open.

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1341 8.5 Open problems

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1343 Several extensions are natural. First, the $\varepsilon \downarrow 0$ limit of regularized Fisher–Rao geometry

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1345 could be studied using tools from singular information geometry. Second, the active-

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1347 eigenspace curvature phenomenon could be attacked symbolically. Third, the leakage

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1349 model could be extended to masked, higher-order, non-Gaussian, or profiled settings.

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1351 Fourth, full-round ciphers would require propagation through linear layers and key

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1353 schedules, which is outside the isolated S-box model considered here.

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1356 9 Conclusion

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1358 We presented a boundary information-geometric framework for the local S-box dis-

1359

1360 tributions that arise in side-channel key recovery. The main modelling point is the

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1362 separation between three spaces: the regular full-support Walsh exponential family,

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1364 the singular key-induced boundary orbit, and the adversary’s belief simplex. This

1365

1366 separation makes Fisher, KL, and projection statements mathematically well scoped.

1367

1368 For bijective S-boxes, Walsh coordinates give closed-form boundary covariance

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1370 matrices. The active covariance subspace has dimension $2^s - 1$ and all active eigenvalues

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1372 equal 2^s in the unnormalised Walsh basis. Thus this part of the geometry is a general

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1374 structural consequence of bijectivity.

1375

1376 For CPA, Bayesian updates occur on the belief simplex. Under Hamming-weight

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1378 Gaussian leakage, the scalar variance information is a baseline fixed by bijectivity,

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1380 while pairwise leakage gaps determine expected posterior-odds growth against wrong

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1382 keys. Computations on PRESENT, GIFT, and SKINNY illustrate this distinction and

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1384 reproduce the active curvature diagnostic $K = -1/4$ in the tested 4-bit cases.

The framework is not a new S-box classification and not a standalone attack. Its contribution is a controlled geometric language for connecting singular S-box distributions, boundary covariance, and side-channel Bayesian inference.

A Reproducibility Details

This appendix records the computational protocol used for the numerical claims in Section 7. The computations are intentionally small and exhaustive: for $s = 4$, all input values, all key hypotheses, all Walsh coordinates, and all active coordinate planes are enumerated.

A.1 Software environment

The distributed scripts use Python 3 and standard numerical packages. The core dependencies are `numpy` for exact finite-array enumeration and linear algebra, and `matplotlib` for figure generation. No randomized optimization, training procedure, or stochastic sampling is used for the reported tables. When a random 4-bit bijection is used in `verify_curvature.py`, the script fixes the pseudorandom seed explicitly.

A.2 Commands

The main numerical outputs can be regenerated with the following commands from the project root:

```
python gen_figures.py
python compute_curvatures.py
python verify_curvature.py
```

The first command regenerates the Walsh-spectrum and local key-separation figures. The second command constructs the boundary covariance matrix, projects onto the

active eigenspace, computes the projected third cumulant contractions, and evaluates the curvature diagnostic. The third command is a sanity check for the value $K = -1/4$ on PRESENT and on an additional fixed random bijection.

A.3 Arithmetic and precision

The Walsh coefficients, boundary covariance entries, and third cumulant entries are obtained from finite sums over $\{0, 1\}^4$. These quantities are therefore integer or rational before diagonalization. The eigenspace projection and the curvature table are evaluated in floating-point arithmetic after the exact finite sums have been constructed. The reported curvature values are therefore reported as reproducible numerical diagnostics for the tested S-boxes.

A.4 Sign conventions

The Walsh transform convention is

$$\widehat{S}(\alpha, \beta) = \sum_{a \in \{0, 1\}^s} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a)}.$$

The active-eigenspace curvature diagnostic uses the convention in (36). Changing the Riemann tensor sign convention would change the sign of all reported sectional curvature values; the scripts and the paper use the same convention.

A.5 Reproducibility coverage

The computational protocol covers the quantities reported in the paper: Walsh spectra, boundary covariance matrices, active ranks/eigenvalues, local spectral key-separation proxies, Hamming-weight leakage descriptors, and the projected active-eigenspace curvature diagnostic for the tested 4-bit S-boxes.

Declarations	1473
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Availability of data and materials	1475
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	1477
The datasets supporting the conclusions of this article are included within the article	1478
and its additional file. The additional file <code>supplementary_material.zip</code> contains the	1479
scripts used to regenerate the numerical tables and figures.	1480
	1481
	1482
Software information: project name, Boundary Information Geometry for S-box	1483
Side-Channel Leakage; project home page, https://github.com/EkodeckStephane/	1484
<code>IG-based-Sbox-SCL</code> ; archived version, not applicable before repository release or DOI	1485
assignment; operating system, platform independent; programming language, Python	1486
3; other requirements, <code>numpy</code> and <code>matplotlib</code> ; license, not specified; restrictions to	1487
use by non-academics, none beyond the repository license status.	1488
	1489
	1490
	1491
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	1493
Competing interests	1494
	1495
The authors declare that they have no competing interests.	1496
	1497
	1498
Funding	1499
	1500
No specific funding was received for this work.	1501
	1502
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